



Samsung's new app

Samsung introduces the Smartphone Gamepad and Mobile Console app.
<http://dgit.in/smsgapp>

Nearby Tweets

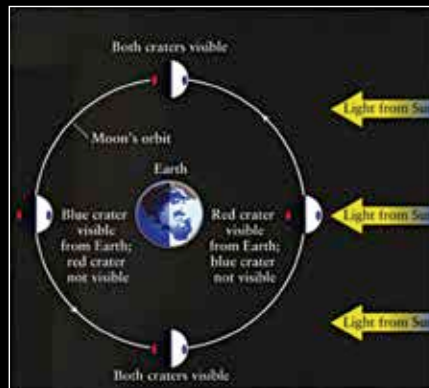
Twitter's new experimental feature shows tweets nearby their location on a map
<http://dgit.in/twtrnrby>

the far side with their own eyes while their vessel orbited the Moon.

By the time Neil Armstrong stepped on the surface numerous photographs and eyewitness observations had already been analysed by scientists. Over time the improvements in technology has made those grainy images a thing of the past. The most recent addition to our knowledge of the far side is from the images taken by the Japanese orbiter Kaguya, that photographed the 185 km crater of Tsiolkovsky Crater with a full HDTV camera.

However despite the quality of the HD images it remains very hard to imagine the Moon as it must appear from space. Even the full Moon isn't truly full and we are left with only half the picture at any given time due to our "tidally locked" orbits. The rotation around its axis, the rising and setting of light across its surface and the real sense of its motion can't be appreciated from mere photographs. Scientist at NASA felt the same way so they did something about it.

In September 2013, NASA used images taken by the Lunar Reconnaissance Orbiter Camera (LROC) over a period of four years to compile a 25 second video clip that shows exactly that - the closest possible vision to observing the Moon from

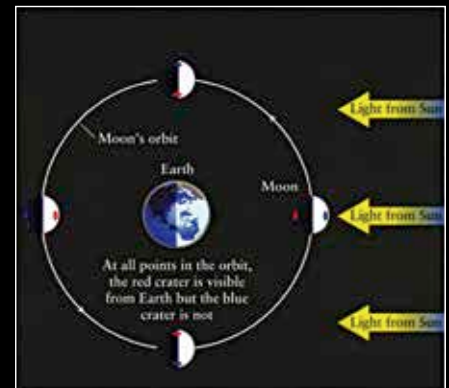


a. If the Moon did not rotate, we could see all sides of the Moon

a position in space. Using the LRO's Wide Angle Camera (WCA), NASA was able to take advantage of the 90 degree field of view to create a to-scale near global topographic map and make a timelapse video.

The challenge of accomplishing this task required a meticulous and innovative strategy devised by NASA scientists. An equation was determined that accounted for the angle of the sun, the viewing angle and reflectance changes and applied to every pixel of the 110,000 wide angle camera shots that the LRO took since it's launch in 2009.

The correction from this composite



b. In fact the Moon does rotate and we see only one face of the Moon

allowed for a crystal-clear view of the dark side of the Moon and demonstrated how even the dark side receives regular sunlight. But its important to note that many impact craters on the Moon don't receive any sunlight at all. The reason for this is simple - the 1.6 degree tilt of the Moon causes sunlight to fall nearly perpendicular to the surface, as compared to Earth's 23.4 degree tilt which ensures that sunlight reaches all surfaces. This difference in relation to the sun results in the Moon having dark shadows in the craters and crevices of its surface.

Moon Origins: More Mysteries Abound

Since 1969, six Apollo missions have allowed humans to walk on the Moon, the last of which was in December 1972 when astronauts Eugene Cernan and Harrison Schmitt became the last humans to walk the lunar surface. The 41 years since have seen very little interest in the Moon, and only recently have other countries like China and India achieved the ability to launch independent space missions.

Scientists now believe that the scientific ambition and legacy of the Moon landings needs to be reignited in order to investigate the fundamental questions relating to the origins of the Moon, which to this day are only theory and conjecture, held back due to insufficient resources. Scientists believe that in order to study the origins of the Moon and the history of our planet's evolution humans need to visit the Moon across its topography.

All manned Moon landings have taken



The resulting debris from the collision between a Mars-sized planet and Earth is theorised to have given birth to a debris field that eventually became the Moon.